INCH-POUND
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

THREAD, COTTON

The General Services Administration has authorized the use of this Commercial Item Description by all federal agencies.

1. SCOPE. This Commercial Item Description (CID), covers the requirements for cotton thread used for sewing various clothing, footwear, and equipage items,

2. CLASSIFICATION. The cotton thread shall conform to the following types:

2.1 Types

Type I - Machine thread, soft finish
Type II - Machine thread, glazed finish
Type III - Deleted
Type IV - Shoe thread, soft finish
Type V - Shoe thread, glazed finish

3. SALIENT CHARACTERISTICS.

3.1 General description. The thread shall be spun from combed cotton and shall have a soft, glazed as specified in the contract or purchase order (see 7.6). The glazing compound may be any commercial available agent intended for this purpose. The final direction of the thread twist shall be "Z" twist.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: DLA Troop Support Standardization Team, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil.

AMSC N/A FSC 8310

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3.2 Physical characteristics and requirements. The thread shall conform to the physical characteristics specified in Tables I, II, and III, when tested in accordance with Table IV.

TABLE I. Physical characteristics for Types I and II.

Tex 1/	Ply	Breaking strength, lbs. (min.)
21	2	1.0
27	2	0.9
30	2 or 3	1.2
30	3	1.5
40	3	1.8
50	3	2.2
60	3	3.0
80	4	3.8
90	4	4.5
105	4	5.3
150	4	6.8
180	3	6.5
210	3 or 4	9.5
270	3,4 or 5	12.5
350	5 or 6	15.5
240	4	10.0
210	3	7.0
270	4	11.0

1/ Tex size based on weight in grams/1,000 meters thread.

TABLE II. Physical characteristics for Type IV and V.

Tex 1/	Ply	Breaking strength lbs. (min.)
270	3 or 4	13.0
350	5 or 6	17.5
400	6	21.0
450	7	24.5
500	8	28.0
600	9	31.5
700	10	35.0
700	11	38.5
800	12	42.0

1/ Tex size based on weight in grams/1,000 meters thread.

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TABLE III. Physical requirements and test methods.

Characteristic	Requirement	Test method
Fiber identification	3.1	ASTM D276 or AATCC 20 (see 7.4)
Twist direction	3.1	ASTM D204
Tex	Table I and II	ASTM D204
Ply	Table I and II	Visual
Breaking strength	Table I and II	ASTM D204
Mildew resistance: 1/ Initial After leaching	 75% of initial breaking strength 75% of initial breaking strength	 AATCC 30, Test 1, 2/ & ASTM D204 3.3.1, AATCC 30, Test 1, 2/ & ASTM D204
Colorfastness : Laundering After three (3) cycles Light (after 40 AFU or 170 kJ/(m^2nm)@420 nm Perspiration (acid & alkaline) Weathering (after 80 AFU) or 340 kJ/(m^2nm)@420 nm	 3-4 (min.) 3-4 (min.) 3-4 (min.) 3-4 (min.)	 AATCC 61, Test 3A, 3/, 4/, 5/ AATCC 16.3, Option 3, 6/ AATCC 15, 4/ AATCC 169, 6/
Knots	3.4	Visual
Labile sulfur	3.5	3.5.1
Visual shade matching	3.6	3.6.1
Finish	3.1	Visual
Toxicity	3.8	3.8.1

1/ When required by contract or purchase order (see 7.6)
2/ A minimum of five (5) specimens shall be tested. The soil shall be made of an equal blend of good topsoil, well rotted and shredded manure, and coarse sand. Samples shall be exposed for 12 weeks.
3/ The specimen must be dried after each cycle.
4/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change and AATCC Evaluation Procedure 2, Gray Scale for Staining.
5/ On the color transfer cloth evaluation, only the stain on the cotton fibers of the color transfer cloth shall be evaluated.
6/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change.

3.3 Mildew resistant treatment. The thread shall have a mildew resistant treatment when specified in the contract and meet the requirements as specified in Table III (see 7.6).

3.3.1 Leaching resistance. Specimens to be tested for mildew and rot resistance shall be submerged in a suitable tank containing water at a temperature of 80°F (+5) and allowed to remain immersed for a period of 24 hours. Water flow through a tube into a tank is adjusted to

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ensure a completed change of water not less than three times in 24 hours. The pH and temperature of the water is recorded and included in the report of the results.

3.4 Knots. Types I and II thread shall average not more than one full thread knot or splice per 1,000 yards. Types IV and V thread shall average not more than one full thread knot or splice per 600 yards.

3.5 Labile sulfur. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid shall not be permitted.

3.5.1 Presence of labile sulfur test. In the determination of presence of labile sulfur in textile materials with lead acetate, two (2) 1.50 (± 0.01) gram samples from each material submitted for evaluation shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. The samples shall be submersed in a stannous chloride solution that contains 100 grams of stannous chloride crystals ACS in 100 milliliters of hydrochloric acid ACS (35 percent concentration) and 50 milliliters of distilled water. A filter paper wet out with a 5.0 percent lead acetate solution shall be placed over top of the test tube. The lead acetate solution contains 5.0 grams of lead acetate CP reagent grade and enough distilled water to make up a 100 milliliter solution; if the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear. The test tube containing the textile sample, stannous chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

3.6 Color. The color of the dyed thread shall be as specified in the applicable end item specification or in the contract order. The color shall be natural, bleached, or dyed as specified in the contract or purpose order (see 7.6).

3.6.1 Visual shade matching. The color and appearance of the finished thread shall match the standard sample when viewed using AATCC Evaluation Procedure 9, Option A, with sources simulating artificial daylight D75 illuminant with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856K (± 200).

3.7 Finished materials. The finished thread shall have no chemical finishes or treatments other than those commonly used or specified (see 3.1) which have been demonstrated to have no harmful effects on the fiber, including effects of prolonged storage. No finish or treatment shall be applied for the purpose of increasing breaking strength. There shall be no noticeable wicking of the treatment on the thread to adjacent material when sewn.

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3.8 Toxicity. The finished thread shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 3.8.1. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.8.1 Toxicity test. When required (see 7.6), an acute dermal irritation study and a skin sensitization study shall be conducted on laboratory animals. When the results of these studies indicate the thread is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 7.2.3). If the toxicity requirement (3.8) can be demonstrated with historical use data, toxicity testing may not be required (see 7.6).

3.9 Put-up. Unless otherwise specified, the thread shall be put-up on holders such as commercial spools, cones, or tubes as specified in the contract. The thread shall be wound around the specified holder in one continuous piece, so each turn and layer is free of entanglement. The outside ending of the thread shall be secured to prevent unwinding, loosening, or slippage during handling, shipping, or storage.

3.10 Labeling. Each holder shall have a label, adhered securely so as to remain in place and be clearly legible until all thread has been removed. The label shall be printed and include information related to length in yards, direction of twist, color, Tex, name of thread manufacturer, and nomenclature specified type.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own specifications, standards, and quality assurance practices, and be the same product offered in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.1.1 Market acceptability criteria. The item offered must have been sold to the Government or commercial market.

6. PACKAGING

6.1 Packaging. Preservation, packing, and marking shall be specified in the contract or order. (See 7.6).

7. NOTES.

7.1 Sources of Government documents.

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7.1.1 EPA Pollutants/toxics subtopics are available online at https://www.epa.gov/regulatory-information-topic.

7.1.2 Federal Acquisition Regulations are available online at http://acquisition.gov/far/index.html

7.2 Sources of Non-Government documents.

7.2.1 AATCC Test Methods are available online at http://www.aatcc.org.

7.2.2 ASTM Standards are available online at http://www.astm.org.

7.2.3 Modified Draize Procedure. Principles and Methods of Toxicology, A Wallace Hayes (editor), are available online at https://www.crcpress.com/.

7.3 Standard sample. For access to standard shade samples of thread, address the contracting activity issuing the invitation for bids or request for proposal.

7.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

7.5 Type III, Machine thread, mercerized finish. Type III has been deleted due to the unavailability of mercerized cotton thread in the USA. Type III is not Berry Amendment compliant.

7.6 Ordering data. The contract or purchase order should specify the following:

a. Title, number, and date of this Commercial Item Description
b. Type, Tex, and ply of thread required (see 2.1, 3.2)
c. Mildew resistance when required (see 3.3)
d. Color required (see 3.6)
e. When toxicity testing is required (see 3.8)
f. Put-up required if other than specified (see 3.9)
g. Product conformance provisions (see 5.1)
h. Packaging requirements (see 6.1)

7.7 Key words.

Clothing
Shoe
Equipage
Machine sewing

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MILITARY INTERESTS:

Custodians:

Army – GL
Navy – NU
Air Force – 11

Reviewer activity:

Army – MD
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FAS

PREPARING ACTIVITY

DLA-CT

(Project: 8310-2019-006)

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